Software Requirements Specification (SRS)

Second-Hand Vehicle Selling Website

1. Introduction

1.1 Purpose

The purpose of this project is to develop a responsive and user-friendly Second-Hand Vehicle Selling Website where users can browse, search, filter, compare, and view details of used vehicles available for sale. The project is designed for frontend developers and focuses on implementing the complete user interface and frontend interactions.

1.2 Scope

The website will provide a realistic vehicle marketplace experience where users can browse vehicles, search and filter listings, view vehicle details, save favorites, compare vehicles, contact sellers, and submit vehicle listings through a frontend form. Backend services, databases, payment gateways, and real authentication are outside the scope of this project. Mock data and browser storage may be used for demonstration.

2. Objectives

1. Create a professional second-hand vehicle marketplace interface.
2. Develop a responsive and user-friendly website.
3. Implement vehicle search, filtering, and sorting.
4. Display vehicle information using reusable UI components.
5. Implement frontend form validation.
6. Provide favorites and vehicle comparison functionality.
7. Create a detailed vehicle information page.
8. Create a seller vehicle listing form.
9. Practice component-based frontend development.
10. Build a portfolio-level real-world web application.

3. Target Users

3.1 Buyers

- Search for vehicles.
- Apply filters.
- View vehicle details.
- Compare vehicles.

- Save favorite vehicles.
- Contact sellers.

3.2 Sellers

- Enter vehicle details.
- Add vehicle images.
- Enter price and location.
- Add a vehicle description.
- Preview and submit a listing.

3.3 Visitors

- Browse vehicles.
- Search vehicles.
- View vehicle details.
- Use basic filters.

4. Home Page Requirements

- Website logo/name.
- Navigation bar.
- Vehicle search section.
- Vehicle category selection.
- Featured vehicles.
- Recently added vehicles.
- Popular brands.
- Why Choose Us section.
- Customer reviews/testimonials.
- Footer.

Suggested navigation: Home, Buy Vehicle, Sell Vehicle, Compare, Favorites, Login/Register.

5. Vehicle Search Requirements

Users should be able to search for vehicles using relevant search criteria.

- Vehicle name.
- Brand.
- Model.
- Location.
- Minimum price.
- Maximum price.

Example: A user can search for Toyota vehicles in Hyderabad with a maximum price of ₹8,00,000.

6. Vehicle Listing Page

The vehicle listing page should display available vehicles using attractive and reusable vehicle cards.

- Vehicle image.
- Brand and model.
- Manufacturing year.
- Price.
- Kilometers driven.
- Fuel type.
- Transmission.
- Location.
- Favorite button.
- View Details button.

7. Vehicle Filtering

Users should be able to narrow down vehicle results using filters.

7.1 Vehicle Type

- Car
- Bike
- SUV
- Sedan
- Hatchback

7.2 Price Range

- Under ₹2 Lakhs
- ₹2–5 Lakhs
- ₹5–10 Lakhs
- ₹10–20 Lakhs
- Above ₹20 Lakhs

7.3 Fuel Type

- Petrol
- Diesel
- Electric

- CNG
- Hybrid

7.4 Transmission

- Manual
- Automatic

7.5 Year

- 2026
- 2025
- 2024
- 2023
- 2022
- Older

7.6 Location

Users should be able to select or enter a city/location.

8. Vehicle Sorting

- Price: Low to High.
- Price: High to Low.
- Newest vehicles.
- Oldest vehicles.
- Lowest kilometers.
- Highest kilometers.

9. Vehicle Details Page

Selecting a vehicle should open a detailed vehicle information page.

9.1 Vehicle Images

- Main vehicle image.
- Multiple thumbnail images.
- Image gallery.

9.2 Vehicle Information

- Brand.
- Model.
- Variant.
- Manufacturing year.

- Registration year.
- Price.
- Kilometers driven.
- Fuel type.
- Transmission.
- Number of owners.
- Vehicle condition.
- Color.
- Location.

9.3 Description

Display a detailed description of the vehicle.

9.4 Features

- Air Conditioning
- Power Windows
- ABS
- Airbags
- Parking Sensors
- Rear Camera
- Bluetooth
- Touchscreen
- Sunroof

9.5 Seller Information

- Seller name.
- Location.
- Contact button.
- Message seller button.

9.6 Actions

- Add to Favorites.
- Compare.
- Contact Seller.
- Share.

10. Favorites

Users should be able to save vehicles for later reference.

- Add or remove vehicles from favorites.

- Change the favorite icon when selected.
- Display saved vehicles on a Favorites page.
- Persist favorites using browser storage.

11. Vehicle Comparison

Users should be able to select vehicles and compare important information side by side.

Feature	Vehicle 1	Vehicle 2
Brand	Toyota	Hyundai
Model	Innova	Creta
Year	2021	2022
Price	₹15 Lakh	₹14 Lakh
Fuel	Diesel	Petrol
Transmission	Automatic	Manual
KM Driven	42,000	30,000

- Add vehicles to comparison.
- Remove vehicles from comparison.
- Display selected vehicles side by side.

12. Sell Vehicle Page

The Sell Vehicle page should provide a form through which sellers can enter their vehicle information.

12.1 Vehicle Information

- Vehicle type.
- Brand.
- Model.
- Variant.
- Manufacturing year.
- Registration year.
- Fuel type.
- Transmission.
- Kilometers driven.
- Number of owners.
- Color.

12.2 Pricing

- Expected selling price.

12.3 Location

- State.
- City.
- Area/Pincode.

12.4 Vehicle Images

Users should be able to select vehicle images for the listing.

12.5 Description

Users should provide a detailed description of the vehicle.

12.6 Contact Information

- Seller name.
- Phone number.
- Email address.

12.7 Actions

- Save Draft.
- Preview Listing.
- Submit Listing.

13. Login Page

- Email/mobile number.
- Password.
- Remember Me.
- Forgot Password.
- Login button.
- Create Account link.

Frontend validation should be provided for the login form.

14. Registration Page

- Full name.
- Email.
- Mobile number.
- Password.
- Confirm password.
- City/location.
- Terms and conditions checkbox.

14.1 Validation

- Required fields.
- Email format.
- Mobile number.
- Password length.
- Password confirmation.

15. User Dashboard

The dashboard should provide a central place for users to manage their frontend account information and activities.

- Profile.
- My Listings.
- Favorites.
- Compared Vehicles.
- Enquiries.
- Edit Profile.
- Logout.

16. Contact Seller

The vehicle details page should provide a contact form.

- Name.
- Email.
- Phone number.
- Message.

After successful submission, display an appropriate success message.

17. Responsive Design

The website must work correctly across desktop, laptop, tablet, and mobile screen sizes.

- Navigation should convert to a hamburger menu on smaller screens.
- Vehicle cards should use an appropriate responsive layout.
- Filters can appear in a modal or drawer on mobile.
- Images should be responsive.
- Buttons should be touch-friendly.
- Comparison tables should support horizontal scrolling where necessary.

18. UI Requirements

The interface should have a clean, modern marketplace design.

- Navbar.
- Footer.
- Search bar.
- Vehicle card.
- Filter sidebar.
- Dropdowns.
- Buttons.
- Modal.
- Image gallery.
- Breadcrumb.
- Form components.
- Alert/toast messages.
- Loading indicator.
- Empty-state messages.
- Pagination.

Example empty-state message: “No vehicles found. Try changing your search filters.”

19. Routing Requirements

The application should support navigation between the major pages and vehicle details.

- /
- /vehicles
- /vehicles/:id
- /sell
- /favorites
- /compare
- /login
- /register
- /dashboard
- /contact

20. Form Validation Requirements

- Required-field validation.
- Valid email format.
- Valid mobile number.

- Valid price input.
- Password length validation.
- Confirm-password validation.
- Clear error messages near invalid fields.

21. Browser Storage Requirements

Browser storage such as localStorage may be used to demonstrate persistence without a backend.

- favoriteVehicles
- compareVehicles
- userData
- sellerListings

22. Non-Functional Requirements

22.1 Performance

- Pages should load quickly.
- Images should be optimized.
- Unnecessary re-rendering should be minimized.

22.2 Usability

- Navigation should be simple.
- Buttons should have clear labels.
- Forms should provide meaningful validation messages.

22.3 Accessibility

- Use proper labels.
- Provide alt text for images.
- Support keyboard-friendly navigation.
- Maintain readable text and layouts.
- Use semantic HTML where appropriate.

22.4 Compatibility

- Google Chrome.
- Microsoft Edge.
- Firefox.
- Safari.

23. Security Considerations

- Do not store real passwords or sensitive credentials in the frontend.
- Do not hardcode sensitive information.
- Validate user inputs.
- Do not collect real payment information.
- Use mock authentication for demonstration purposes.

24. Error Handling

- No vehicles found.
- Invalid vehicle ID.
- Form validation errors.
- Empty favorites.
- Empty comparison list.
- Failed image loading.
- Invalid search input.

Example: “Vehicle Not Found – Sorry, the vehicle you are looking for is no longer available.”

25. Expected Pages

11. Home
12. Vehicle Listing
13. Vehicle Details
14. Sell Vehicle
15. Favorites
16. Compare Vehicles
17. Login
18. Register
19. User Dashboard
20. Contact Seller
21. 404 / Page Not Found

26. Suggested Project Structure

- src/
- ├── components/
- │ ├── Navbar
- │ ├── Footer
- │ ├── SearchBar
- │ ├── VehicleCard

- │ ├── FilterSidebar
- │ ├── VehicleGallery
- │ ├── SellerCard
- │ └── Pagination
- ├── pages/
- │ ├── Home
- │ ├── Vehicles
- │ ├── VehicleDetails
- │ ├── SellVehicle
- │ ├── Favorites
- │ ├── Compare
- │ ├── Login
- │ ├── Register
- │ └── Dashboard
- ├── data/
- │ └── vehicles.json
- ├── context/
- │ └── VehicleContext
- ├── App
- └── main

27. Sample Vehicle Data

The project should use mock vehicle data for demonstrating search, filtering, sorting, favorites, and comparison.

- id: 1
- type: Car
- brand: Toyota
- model: Innova Crysta
- year: 2021
- price: 1550000
- fuel: Diesel
- transmission: Automatic
- kilometers: 42000
- owners: 1
- location: Hyderabad
- color: White
- image: /images/innova.jpg

At least 15–20 vehicle records should be included so that search, filtering, sorting, favorites, and comparison can be demonstrated properly.

28. Project Workflow

28.1 Buyer Workflow

Home → Search Vehicle → Apply Filters → Vehicle Listing → Vehicle Details → Add to Favorites / Compare → Contact Seller

28.2 Seller Workflow

Home → Sell Vehicle → Enter Vehicle Details → Add Images → Enter Price & Location → Preview Listing → Submit Listing

29. Frontend Project Deliverables

- Complete source code.
- All required pages and routes.
- Reusable UI components.
- Mock vehicle dataset.
- Responsive desktop, tablet, and mobile layouts.
- README documentation containing project description, features, setup instructions, and screenshots.

30. Minimum Project Requirements

- Responsive homepage
- Navigation bar
- Vehicle search
- Vehicle listing cards
- Search filtering
- Sorting
- Vehicle details page
- Image gallery
- Favorites functionality
- Vehicle comparison
- Sell vehicle form
- Login page
- Registration page
- Form validation
- Browser storage

- Responsive mobile design
- Routing
- Mock vehicle data
- 404 page

31. Future Enhancements

- Real user authentication.
- Database integration.
- Admin dashboard.
- Real vehicle listings.
- Seller verification.
- Online payment.
- Buyer-seller chat.
- Email/SMS notifications.
- Vehicle document verification.
- Location/map integration.
- AI-based vehicle price prediction.
- Personalized vehicle recommendations.

32. Conclusion

The Second-Hand Vehicle Selling Website is a practical frontend development project that gives students experience in building a real-world marketplace application. The project covers responsive design, reusable components, routing, forms, validation, search, filtering, sorting, browser storage, comparison, favorites, and user-friendly interfaces. Completing this project will provide students with a strong portfolio project demonstrating their ability to build a complete modern web application from the frontend side.